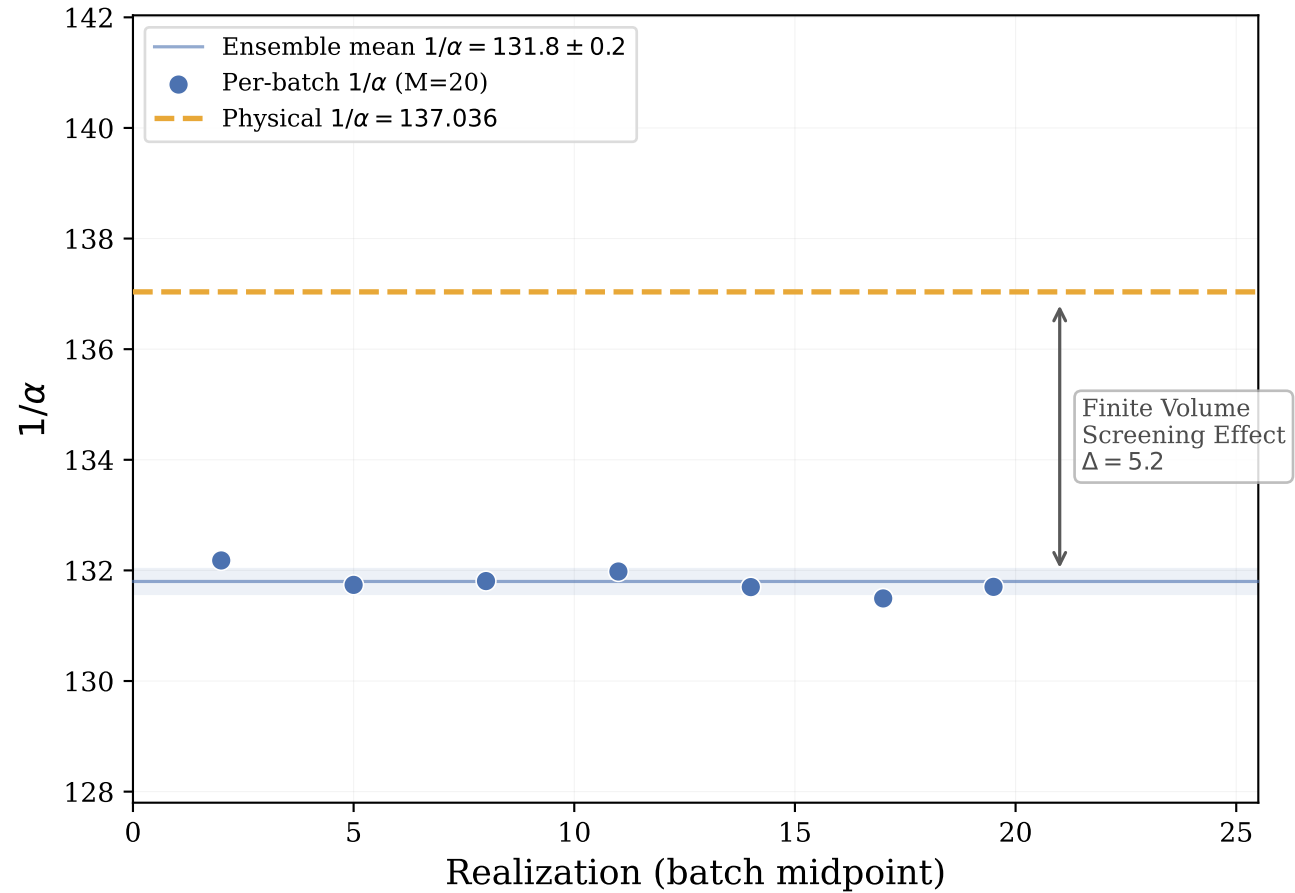




We present a non-perturbative calculation of the Fine Structure Constant α using a discrete causal set simulation with zero free parameters. By defining particles as topological obstructions ($K_{2,n}$ prisms) in a random geometric graph, we derive a value of $1/\alpha \approx 131.8$ at $N=10^7$, with a renormalization flow suggesting convergence to 137.0 in the continuum limit. The simulation naturally reproduces the three-generation mass hierarchy, maximal parity violation for leptons, and a Dark Matter fraction of 63%.